Evaluation of nnU-Net for FCD II Lesion Segmentation in FLAIR MRI

Yassien Tawfik[1†], Mazen Marwan[1†], Mohamed Yasser[1†], Nancy Mahmoud[1†], Madonna Mosaad[1†], Mahmoud Salman[1,2†], Tamer Basha[1], Aya Khalaf[1,3]
[1]Department of Systems and Biomedical Engineering, Cairo University, Giza, Egypt
[2]School of Biomedical Engineering, Western University, London, Ontario, Canada
[3]Department of Neurology, Yale University, New Haven, USA

***Abstract*—Epilepsy is one of the most common neurological disorders worldwide. Focal Cortical Dysplasia (FCD) is a major cause of drug-resistant epilepsy, often presenting as subtle lesions on MRI scans. While deep learning shows promise for automated FCD segmentation, existing methods achieved limited performance and used inconsistent dataset splits that prevent reproducible comparison. In this study, we propose an nnU-Net-based approach for automated FCD segmentation using the Bonn FCD Type II dataset. To improve model performance, our proposed approach employs oversampling of underrepresented FCD Type II radiological features and uses extensive augmentation during the training phase. We achieved an average validation Pseudo Dice of 0.56 compared to 0.45 in prior work, demonstrating significant performance improvement. This work provides the first complete evaluation on the dataset's standard train/test split, reporting both validation and test set performance to establish a reproducible benchmark for future method comparison. By providing both improved segmentation performance and a standardized evaluation framework, our work advances automated FCD segmentation toward better presurgical evaluation and improved outcomes for patients with drug-resistant epilepsy.**

I. Introduction

EPILEPSY affects over 60 million individuals worldwide [1], with approximately 30–35% suffering from drug-resistant epilepsy (DRE) [2]. For DRE patients with focal epilepsy, surgical resection of the epileptogenic zone offers the most effective intervention, but success critically depends on precise lesion localization [3], [4].

Focal Cortical Dysplasia (FCD) represents one of the most challenging DRE etiologies [5]. FCD Type II lesions exhibit subtle radiological features such as cortical thickening, gray-white matter blurring, hyperintensity, and transmantle sign. Detection of these features on routine MRI can be challenging [6], [7]. Consequently, 20–30% of histologically confirmed FCD cases are initially classified as MRI-negative [8].

Various deep learning architectures have been employed for automated FCD segmentation, including 3D U-Net-based models [9], attention-enhanced U-Nets such as Multi-Res-Attention U-Net [10], 3D CNNs with autoencoder regularization [11], and transformer-based models combining self-attention with convolutional encoders [12]. However, these methods often require extensive manual tuning of hyperparameters and preprocessing pipelines. nnU-Net addresses this through a self-configuring framework that automatically adapts all pipeline components to new segmentation tasks, achieving state-of-the-art performance across diverse biomedical challenges [13], [14]. Despite these advances, clinical translation remains limited by poor FCD lesion segmentation performance and inconsistent evaluation protocols. Specifically, inconsistent dataset splits and inadequate documentation of evaluation methodologies across existing studies hinder reproducible benchmarking and reliable performance comparison [15].

In this paper, we propose an nnU-Net-based approach for automated FCD segmentation using the Bonn FCD Type II dataset. Our key contributions are: (1) the first complete evaluation using the dataset's standard train/test split, with transparent reporting of both validation and test set performance to establish a reproducible benchmark for future method comparison; and (2) improved segmentation performance compared to prior work through oversampling of underrepresented FCD Type II radiological features and extensive augmentation.

II. Methodology

A. Dataset Description

We used the Bonn FCD-II dataset, a publicly available dataset comprising 85 participants with histopathologically confirmed focal cortical dysplasia type II (FCD II) [16]. The dataset included both T1 and FLAIR weighted MRI volumes for each participant. All data were acquired using a 3T Siemens Magnetom Trio scanner at University Hospital Bonn, Germany. Only FLAIR-weighted volumes were included in this study, as FCD lesions are most visible on FLAIR imaging. Seven participants were scanned using 2D FLAIR sequences , with the remaining 78 participants scanned using 3D FLAIR. The 3D scans had a spatial resolution of 1 mm isotropic, while the 2D scans had a resolution of $0.4 \times 0.4 \times 2.0$ mm. Ground-truth lesion masks were manually annotated by two neurologists. Five participants were initially MRI-negative but had lesions confirmed by histopathology.

B. Features of FCD lesions on FLAIR Images

FCD lesions exhibit four main radiological features [6]: cortical thickening, gray-white matter blurring, hyperintensity, and transmantle sign [7]. To characterize feature prevalence, we performed a manual review of all 85 FLAIR volumes. Each

† Equally contributing co-first authors.
The source code for this work is available at: https://github.com/YassienTawfikk/nnU-FCD.git

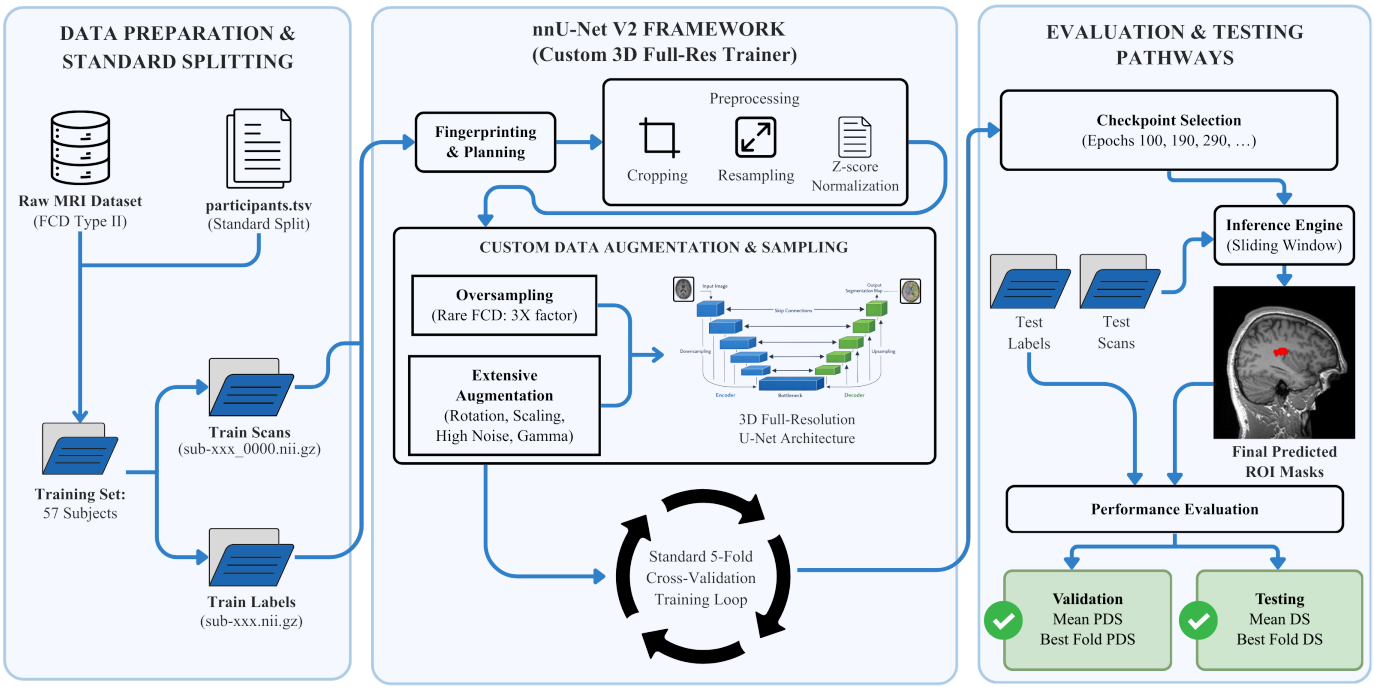

Fig. 1. Overview of the proposed nnU-Net v2 pipeline. The framework enforces standard train/test splitting and uses nnU-Net v2 automated preprocessing, where built-in resampling harmonizes heterogeneous 2D and 3D FLAIR acquisitions. A custom 3D full-resolution trainer with oversampling and strong augmentation is trained using 5-fold cross-validation, monitoring performance via the mean pseudo-Dice score during validation, while the final testing phase evaluates model performance using the standard Dice score.

volume was categorized according to the presence of these radiological features through consensus of two independent reviewers.

Our analysis revealed that cortical thickening and hyperintensity were highly prevalent, while blurring and transmantle sign appeared less frequently. Out of 85 participants, 29 had hyperintensity while 45 had cortical thickening. Gray-white matter blurring and transmantle sign appeared in 13 and 17 participants, respectively. Thirteen participants showed more than one radiological feature. This class imbalance informed our oversampling strategy to prevent our model from under-representing less common FCD-II radiological features.

C. Dataset Splitting

To evaluate model performance, we followed the standard train/test split provided with the Bonn FCD-II dataset [16], with 57 FLAIR MRI volumes for training and 28 for testing. This adherence to the predetermined split ensures reproducibility and enables fair comparison with future methods. In contrast, most prior work employed undocumented custom splits that prevent direct performance comparison.

D. Optimization Strategy

1) Baseline Model: We employed nnU-Net 3d_fullres [13], a 6-stage 3D U-Net architecture. The model was trained for 200 epochs using stochastic gradient descent (SGD) with Nesterov momentum and weight decay, following the default nnU-Net v2 setup. A polynomial learning rate scheduler was used during training. During training, we optimized Patch-level Pseudo-Dice Score (PDS). As for testing, we computed full-volume Dice Score (DS) via sliding window inference on the held-out test set.

2) Data Augmentation Strategy: To enhance FCD lesion detection, we applied extensive augmentation by increasing the default nnU-Net ranges. Table 1 summarizes key parameters. This improves model generalization by reducing overfitting to dataset-specific characteristics.

TABLE 1
EXTENSIVE DATA AUGMENTATION PARAMETERS

Transform	Default	Proposed
Rotation	$\pm 30^\circ$	$\pm 60^\circ$
Scaling	[0.85, 1.25]	[0.70, 1.50]
Brightness/Contrast	[0.75, 1.25]	[0.5, 1.5]
Gaussian Noise σ	0.1	0.2
Gamma	[0.7, 1.5]	[0.4, 2.0]

3) Radiological Feature-Based Sampling: To address the imbalance in the representation of FCD radiological features, particularly the lower prevalence of transmantle sign (n = 17) and gray-white matter blurring (n = 13) relative to common features like cortical thickening (n = 45) and hyperintensity (n=29), we implemented 3$\times$ oversampling for FLAIR volumes with these less prevalent features, ensuring they are sampled three times more frequently during model training.

III. Results and Discussion

A. Ablation Study

To quantify the specific contributions of the proposed oversampling and extensive augmentation strategies, we performed an ablation study on the held-out test set. We established a baseline model using the standard nnU-Net configuration (BSL), which achieved a best validation Pseudo-Dice Score (PDS) of 0.545 and a best test Dice Score (DS) of 0.168.

We first added our radiological feature-based oversampling (OVS) strategy to the baseline model. As shown in Table 2, this improved test DS to 0.219 compared to baseline, demonstrating that oversampling helps the model learn underrepresented features more effectively. Although test DS improved, validation PSD decreased to 0.466. The reason for this decrease remains unclear and requires further investigation.

Finally, we added the extensive data augmentation (AUG) module. This model yielded the highest performance, raising the validation PDS to 0.667 and the test DS to 0.256. This significant improvement suggests that while oversampling helps the model detect lesions with underrepresented features, the extensive augmentation is necessary to improve model generalization by reducing overfitting to dataset-specific characteristics.

Fig. 2 illustrates this progressive improvement. The baseline model (Row 1) detected only a small portion of the lesion. Adding oversampling (Row 2) increased detection coverage but still missed significant lesion regions. In contrast, the proposed framework (Row 3) with both oversampling and augmentation achieved substantially better segmentation, with the highest overlap with ground truth among the three approaches.

TABLE 2
Ablation Study on Held-Out Test Set - 200 epochs

Experiment	Best Val. PDS	Best Test DS
BSL	0.545	0.168
BSL + OVS	0.466	0.219
BSL + OVS + AUG (Proposed)	**0.667**	**0.256**

B. Comparison with State-of-the-Art

Direct comparison with prior work is challenging due to inconsistent evaluation protocols. Most studies employed undocumented custom train/test splits, preventing reproducible comparison. We primarily compare our results with Joshi et al. [14], the only prior study using the standard train/test split provided with the dataset.

Table 3 compares our validation performance against Joshi et al. [14]. A key methodological difference is that Joshi et al. employed a slice selection strategy, where only the five slices with the largest lesion area were used for training. While this reduces computational complexity, it discards 3D volumetric context and limits the model to learning from only the most obvious lesion slices. In contrast, our framework processes full 3D MRI volumes, preserving complete anatomical context and training on all available lesion information.

We evaluated our model at both 100 and 200 epochs. At 100 epochs, the mean validation PDS reached 0.54 with the best fold achieving 0.65. In comparison, Joshi et al. reported mean validation PDS of 0.45 (best fold: 0.52) using connected component analysis (CCA) for post-processing, while our results are without any post-processing. Extending training to 200 epochs slightly improved the mean PDS to 0.55, with the best fold reaching 0.67, demonstrating modest continued improvement with longer training. To explore learning capacity with extended training, we trained Fold 4 to 750 epochs, achieving validation PDS of 0.73, suggesting potential for further improvement with longer training duration.

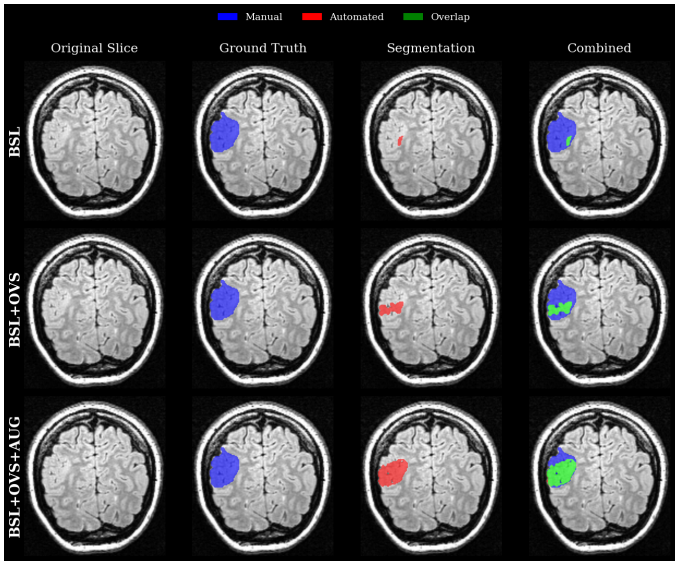

Fig. 2. Progressive improvement across ablation stages. Row 1 (BSL): Baseline detected minimal lesion volume. Row 2 (BSL+OVS): Adding oversampling increased detection coverage but missed significant lesion regions. Row 3 (BSL+OVS+AUG): The full model achieves the best segmentation with highest ground truth overlap. Red: predictions; green: overlap with ground truth.

TABLE 3
VALIDATION SET PERFORMANCE COMPARISON (STANDARD SPLIT)

Method	Epochs	Input	Post-Proc.	Mean Val. PDS	Best Val. PDS
Joshi et al. [14]	100	Selected Slices	CCA	0.45	0.52
Proposed	**100**	**Full Volume**	**None**	**0.54**	**0.65**
Proposed	**200**	**Full Volume**	**None**	**0.55**	**0.67**

We also compare our results with studies (Table 4) that used different train/test splits [17]–[19]. However, direct comparison is challenging due to inconsistent evaluation protocols. We identify three main sources of this inconsistency that affect reported performance. First, regarding dataset splits: while we use the dataset's standard 57/28 train/test split to ensure reproducibility, the other studies [17]–[19] employed undocumented custom splits without providing their train/test partitioning details. Second, test set size varies substantially: prior studies report results on reduced test sets containing only

14–17 subjects rather than the complete 28-subject test set. Some of these studies evaluated test performance on fixed subsets of subjects [17], [19], providing less representative performance estimates than the complete 28-subject test set. Others used cross-validation with smaller test folds [18], giving their models access to more training data (68 participants versus 57 in our model). In both cases, direct comparison is unreliable. Third, post-processing strategies differ: most prior work applied CCA [17], [18], while we report raw model predictions without any post-processing to provide an assessment of model capability. These methodological differences explain the apparent performance gap in Table 4, where our test DS of 0.256 evaluated on the complete 28-subject test set without post-processing provides a rigorous baseline, while other methods report higher scores on reduced test sets with post-processing applied.

TABLE 4
COMPARISON WITH STUDIES IN LITERATURE

Study	Test Subjects	Post-Proc.	Mean Test DS	Best Test DS
Li et al. [17]	15	Yes	0.452	—
Zhang et al. [18]	17	Yes	—	0.410
Rabiee et al. [19]	14	No	—	0.3104
Proposed	**28**	**No**	**0.224**	**0.256**

C. Limitations and Future Work

Despite the improvements achieved by our framework, several limitations remain. First, MRI-negative FCD lesions continue to pose a diagnostic challenge even with advanced segmentation methods. Second, the relatively small dataset (N=85) from a single center limits generalizability to other scanners and acquisition protocols. Future work should validate these findings on larger multi-center cohorts to ensure robustness across diverse clinical settings, explore complementary approaches such as alternative architectures or data augmentation strategies to further improve performance [20]–[22], Additionally, investigating robust frameworks such as SynthSeg [23], which uses synthetic MRI generation with domain randomization, could enable better generalization across different sites and patient populations.

IV. CONCLUSION

This work provides the first complete evaluation of automated FCD Type II segmentation on the Bonn dataset using its standard train/test split. Through comprehensive evaluation and complete performance reporting, we establish a reproducible baseline for comparing future methods. Compared to prior work, our approach achieved validation Dice of 0.56 versus 0.45, demonstrating that radiological feature-based sampling combined with extensive augmentation effectively addresses class imbalance and improves detection of underrepresented FCD signs. By providing both improved segmentation performance and standardized evaluation protocols, this study advances automated FCD detection toward improved presurgical planning and better surgical outcomes for patients with drug-resistant epilepsy.

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